

RESEARCH ARTICLE

Comparative assessment of automated and manual monitoring in comprehensive plant–pollinator communities

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Abstract

1. Pollinator declines pose a significant threat to ecosystem services, making effective monitoring methods critical for conservation efforts. Current research on pollination interactions remains constrained by traditional methods such as direct observations, which have limited spatial and temporal coverage and are inherently biased toward diurnal interactions. Moreover, the presence of human observers can alter pollinator behaviour, and field notebooks serve as the only permanent records, restricting data accessibility and reproducibility.
2. To overcome these challenges, we developed an Automated Camera System (ACS) that integrates Raspberry Pi hardware with YOLOv5, a deep learning-based object detection model, to detect pollinators in plant communities. The system was trained on video recordings of plant–pollinator interactions collected over 4 years across four different islands. Detected pollinators were subsequently classified to the species level by an entomologist, who also determined which individuals had landed on floral structures. We also conducted 13 field campaigns across six study sites over two spring seasons to monitor all flowering plants using both direct observations and multiple ACS units. This approach allowed us to compare the plant–pollinator data acquired by each method.
3. The resulting datasets derived from ACS and direct observations shared similar characteristics (e.g. *species richness* and *Shannon diversity*) and core interactions. However, ACS recorded higher pollinator visitation rates per individual and species than direct observations, likely due to the absence of human observers. Additionally, ACS detected a greater proportion of interactions, particularly low-frequency ones, which significantly influenced network metrics and lead to higher *connectance* and *specialisation* and lower *robustness*. Despite these advantages, ACS had difficulties in detecting small-bodied pollinators (<5 mm), highlighting an area for future refinement.
4. This is the first study to introduce an open-source tool for automatically detecting both nocturnal and diurnal plant–pollinator interactions across natural plant communities and to cross-validate it with direct observations. We foresee that

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the proposed system has broad applications in research and conservation, providing valuable insights into pollination networks across diverse species and ecosystems.

KEYWORDS

deep learning, network metrics, nocturnal pollination, plant–pollinator interactions, Raspberry Pi

1 | INTRODUCTION

The ongoing global decline in biodiversity is having far-reaching effects that extend beyond the extinction of species and populations (Chase et al., 2020; Johnson et al., 2017; Wagner, 2020). This decline is disrupting crucial plant–animal interactions (Traveset & Richardson, 2014), such as pollination (Burkle & Alarcón, 2011; Memmott et al., 2007; Morton & Rafferty, 2017). Pollinators play a vital role in terrestrial ecosystems by facilitating the reproduction of wild and crop plants around the globe (Ollerton et al., 2011). Despite their ecological importance, pollinating insects are increasingly threatened by human-induced environmental changes. More than half of the planet's insect species, including many critical pollinators, are experiencing rapid declines, with nearly one third facing imminent extinction due to anthropogenic pressures such as habitat destruction, invasive species and climate change (Artamendi et al., 2025; Sánchez-Bayo & Wyckhuys, 2019; Simmons et al., 2019; Wagner, 2020). Protecting plant–pollinator networks requires integrating ecological theory, network science and conservation biology to assess how species interactions contribute to ecosystem stability (Bascompte & Scheffer, 2023) and, in turn, to design effective conservation strategies (Ollerton, 2017; Tur et al., 2013).

Meeting this challenge requires new tools and approaches to monitor pollination dynamics more effectively and at scale. Drawing meaningful conclusions effective for conservation policies remains challenging, for the amount of data collection effort, particularly when monitoring pollination dynamics across large spatial and temporal scales. The need for accurate, efficient and scalable methodologies for continuous data collection is more crucial than ever (Breeze et al., 2021; Lamba et al., 2019). Automated monitoring systems offer a promising solution by enabling continuous, high-resolution biodiversity assessments across diverse environments (Høye et al., 2025; Roy et al., 2024). By integrating these systems with plant–pollinator network analyses, researchers can improve interaction resolution, detect shifts in pollination interactions and provide early-warning signals for pollinator decline and habitat degradation. Traditional human-based observations, while valuable, are labor-intensive, limited in scale, require high expertise and are challenging to implement consistently—especially in remote or difficult-to-access locations. Other sampling methods, such as traps or netting (Chinga et al., 2024; Nielsen et al., 2011), have limitations in directly revealing the role of insects in pollination, often requiring statistical analysis to infer 'interactions'. A combination of different methods is typically recommended (Nielsen et al., 2011) for a more comprehensive

census, as individual techniques may under- or oversample specific insect groups, potentially introducing bias into the resulting analyses. Recent advancements in portable, cost-effective and programmable monitoring devices, integrated with deep learning algorithms, have the potential to enhance and extend traditional monitoring approaches, improving our ability to track pollination networks and biodiversity changes at unprecedented scales (Webb, 2018).

In ecological studies, interaction events such as seed dispersal and predation are often documented using camera traps to complement direct observations (O'Connell et al., 2011). However, while motion-sensing camera traps are effective for monitoring vertebrates, they are poorly suited for capturing insect pollinators interacting with flowers. The small size and subtle movements of insects are often indistinguishable from flower movements caused by light winds, leading to detection errors (Høye et al., 2021; Pegoraro et al., 2020). Although the community has conducted several experiments on automatic monitoring in controlled environments, such as gardens (Naqvi et al., 2022), as well as in natural settings, several challenges still limit its widespread implementation. For instance, Steen (2017) effectively demonstrated that digital cameras could capture plant–pollinator interactions and visitation frequencies; however, their method lacks automation, relying instead on manual video analysis and restricting applicability primarily to short-statured, herbaceous plants. In contrast, Sittinger et al. (2024) developed an automated, solar-powered pipeline integrating algorithms for detecting diurnal pollinators, yet their system monitors only pollinator abundances without recording plant–pollinator interactions or addressing observer-related biases. Similarly, Bjerger et al. (2022, 2024) introduced a real-time monitoring system capable of tracking pollinator activity on specific plant species (*Sedum* sp. and *Calluna vulgaris*). While effective within homogeneous backgrounds and low-growing plants, this approach is less suitable for structurally complex plant communities. Additionally, there is a notable gap in the literature regarding effective nocturnal pollination monitoring, with very few existing studies. Notable exceptions include Bjerger et al. (2021) and Roy et al. (2024), who developed automated systems for nocturnal moth detection without specifically studying interactions between pollinators and plants. Similarly, Shibata and Kudo (2023, 2025) utilised time-lapse cameras with flash photography, capturing images at 1-min intervals rather than providing continuous video observations.

According to the current state-of-the-art and the lack of end-to-end automated systems (from scheduling recording to generating the final plant–pollinator interaction network, including the detection, identification and classification of both plants and pollinators)

for pollinator monitoring, most pollination studies still heavily rely on traditional methods, such as direct human observations and entomological collections (Hervías-Parejo et al., 2023; Potts et al., 2006), which come with inherent biases and limitations. First, they are inherently biased toward diurnal data due to the constraints of human observation schedules (Skaldina & Blande, 2025). Second, collecting pollinators from already-declining populations may further contribute to the pollinator crisis (Hallmann et al., 2017). Third, human presence can alter pollinator behaviour, influencing interaction frequencies and dynamics (Skaldina & Blande, 2025). Additionally, in direct observation protocols, field notebooks serve as the only permanent record, making transcription accuracy crucial. Real-time note-taking presents challenges, as observers may struggle to record all interactions without missing key details. Given these limitations, there is a pressing need for new methodologies that enable the accurate, long-term documentation of plant–insect interactions, allowing researchers to review data when needed (Bjerge et al., 2024). These approaches must be not only accurate and scalable but also cost-effective and minimally intrusive to ensure widespread applicability in conservation and ecological research (Høye et al., 2025).

Nowadays, the increasing availability of video monitoring equipment offers a powerful tool for rapidly collecting data on flower visitation. When properly configured, these systems enable researchers to capture and analyse pollination interactions in nearly real time, as well as through subsequent frame-by-frame reviews, facilitating accurate species identification and behavioural analysis (Sittinger et al., 2024). Moreover, advancements in computing power and analytical tools have greatly enhanced our ability to manage and interpret these large datasets (Bjerge et al., 2024; Høye et al., 2021; Keitt & Abelson, 2021). The combination of deep learning algorithms, such as Convolutional Neural Networks (CNNs) (Alzubaidi et al., 2021), along with sensor miniaturisation and wireless communication, is paving the way for continuous, fine-scale ecosystem surveillance. These advancements hold great promise for biodiversity monitoring, improving both detection accuracy and predictive modelling (Bjerge et al., 2024; Borowiec et al., 2022; Christin et al., 2019a; Pettorelli et al., 2025). A widely adopted example is the You Only Look Once (YOLO) model (Redmon et al., 2016), which enables real-time object detection by analysing entire images in a single pass. This efficiency makes YOLO particularly well suited for ecological monitoring, including the detection of insect pollinators. However, challenges remain, as insects can be occluded by vegetation, leading to false detections across consecutive frames. Despite these limitations, deep learning-based approaches have already proven highly effective in species identification from images, videos and even acoustic recordings (Høye et al., 2021, 2025). Deep learning algorithms also play a key role in automating biodiversity assessments, supporting citizen science initiatives and complementing direct observations during fieldwork (Montero-Castaño et al., 2022; Norouzzadeh et al., 2018; Van Klink et al., 2022). These models can now operate on-site through portable devices such as single-board computers (e.g. Raspberry Pi), making automated monitoring more cost-effective and accessible for remote ecological studies (Bjerge et al., 2024; Li

et al., 2023; Sittinger et al., 2024; Steen, 2017). These technologies have significantly advanced plant–insect interaction monitoring, enabling researchers to track insect presence, behaviour and pollination interactions with greater precision (Bjerge et al., 2022, 2024; Jocher et al., 2020; Pegoraro et al., 2020). Despite these advances, no existing methodology integrates both hardware and software into a unified system capable of monitoring diurnal and nocturnal plant–pollinator interactions across an entire community. Current approaches also struggle to account for variation in plant structures and background heterogeneity, limiting their applicability across diverse habitats. In addition, significant challenges in detecting small pollinators are expected (Tariq & Javed, 2025). Furthermore, a formal comparative analysis between datasets obtained from automated systems and those collected through focal observations has yet to be conducted. Addressing these gaps is crucial for developing a more robust, standardised and scalable approach to automated pollination monitoring, ensuring its accuracy and reliability in capturing species interactions across complex ecological landscapes.

In this study, we followed a multi-step approach to develop, implement and validate an open-source Automatic Camera System (ACS) for large-scale monitoring of plant–pollinator interactions in natural field conditions. First, we designed and built the ACS using low-cost, widely available components, enabling scheduled recording during both day and night. Second, we deployed the ACS across four islands over 4 years, generating a dataset consisting of videos of pollination interactions across a variety of plant species and complex backgrounds. Third, we developed a pollinator detection algorithm trained on videos from these field campaigns and also open-source datasets to enhance its robustness. The goal was to create a reliable tool for reducing the time required to review video footage, functioning similarly to a camera trap. Each pollinator detected in videos from Cabrera Island was subsequently reviewed by an entomologist specialised in the island's insect fauna to provide species-level classification and to ensure that the observed interactions were likely associated with flowers. This approach helped avoid missing any true interactions while allowing us to discard incidental flyby insects. Fourth, to compare the ACS data acquisition system with traditional methods, we conducted focal observations on one of the selected islands (Cabrera Island) to build community-level plant–pollinator networks resolved to the species level. These networks were then compared with those derived from the ACS video recordings. This comparison enabled us to assess species diversity and network-level metrics, as well as to identify potential biases between the two methodologies.

Our results demonstrate that the unobtrusive nature of ACS enables the detection of a higher number of interactions per individual pollinator—including both insects and vertebrates—compared to direct human observations. Additionally, ACS captures unique plant–vertebrate pollination events that may be overlooked by traditional sampling methods. Owing to its capacity for extended and continuous monitoring, ACS proves more effective in detecting low-frequency (i.e. rare) interactions that are likely to be missed during brief observational surveys. However, we also identified several

limitations of ACS, particularly in detecting small-bodied insect pollinators due to pixel size constraints, which could lead to their underrepresentation in the video data.

To our knowledge, this is the first study to integrate camera-based monitoring and deep learning for community-level pollination research in natural settings. A key strength of this work lies in the extensive data collection, validated through expert-guided insect identification and direct observations, complemented by targeted insect sampling and ACS censuses. Moreover, the study combines the development, implementation and field application of an ACS originally designed for environmental monitoring, with deep learning algorithms for insect detection and quantitative analysis of plant-pollinator networks. By comparing ACS data with standard focal observations, we provide a robust assessment of network diversity and interaction metrics across entire communities. This approach highlights the strengths and limitations of each method and illustrates the potential of automated tools to improve ecological monitoring. We believe such studies are essential for accelerating the adoption of automated technologies, refining ecological methodologies and advancing future research on plant-pollinator interactions.

2 | MATERIALS AND METHODS

2.1 | Hardware device

The ACS unit illustrated in [Figure 1a](#) comprises seven key components: (1) A system on Chip: Raspberry Pi 4 (Raspberry Pi Foundation, Cambridge, UK); (2) Storage: A 64GB, 64-bit SD card; (3) A Camera Module. For image acquisition, we experimented with three Camera Modules, each offering different characteristics: (a) Camera Module V2 (ACS-V2): A 5-megapixel camera featuring an Omnivision OV5647 sensor and a programmable motorised autofocus for precise focus adjustments; and (b) High-Quality Camera (ACS-HQ): Equipped with a 12.3-megapixel Sony IMX477R stacked, back-illuminated sensor, CS mount and a 25 mm lens with manual focus capability. Additionally, for nocturnal pollinator acquisition, we equipped the unit with the (c) NoIR Camera (ACS-NoIR), which features a 5-megapixel Omnivision OV5647 sensor, a 3.6 mm manual focus lens and two infrared LED torches to capture nocturnal interactions. The ACS is powered by a (4) 10,000mAh Li-polymer rechargeable battery that supports up to 10h of continuous monitoring. Finally, for practical use of the ACS, we designed a (5) two-part 3D case. The inner case holds all hardware components and is suitable for use in favourable weather conditions. The outer case provides a waterproof shell, offering additional protection for the components against environmental conditions. The front of the external case is equipped with a threaded cap, enabling manual adjustment of the camera lens when needed. On the back, we included a power button to start the Raspberry Pi and an HDMI interface for connecting a screen (6) without having to disassemble the case. All watertight seals were made from synthetic rubber. Both cases can be mounted on a standard-thread tripod (7), ensuring stable positioning and ease of transport.

2.2 | Study sites description

The ACS units for automated data acquisition were deployed across four islands in the Balearic archipelago (western Mediterranean Sea): Menorca ($\approx 702 \text{ km}^2$) in 2021, Eivissa ($\approx 571 \text{ km}^2$) and Formentera ($\approx 83 \text{ km}^2$) in 2024 and Cabrera ($\approx 16 \text{ km}^2$) during 2022–2023. In parallel, direct observations were conducted at the same study sites on Cabrera during the same period. See [Figure 1b](#) for an overview of the selected islands and methodology applied.

These Balearic Islands share a characteristic Mediterranean climate, with vegetation across all islands dominated by *Pinus halepensis* and *Pistacia lentiscus*. The landscape is structurally defined by pine forests in inland areas and shrub-dominated communities in coastal dune and rocky systems, mostly established on calcareous soils. The most representative natural plant communities include pine forests, rocky coastal shrublands and coastal dune shrublands. Dominant entomophilous plant species vary by habitat: *Cakile maritima* is typical of coastal dunes; *Cistus monspeliensis* is most representative in pine forests; and *Euphorbia dendroides* and *Limonium* spp. are characteristic of rocky coastal areas.

Six 400 m^2 study sites were established. Two replicates each of coastal dune shrublands, pine forests and rocky coastal shrublands, to capture habitat-level variation (see [Figure 1b](#) for study site locations). Compared to Cabrera, the larger islands—Menorca, Eivissa and Formentera—supported greater plant and pollinator diversity, likely due to their larger area. This broader species pool enabled the construction of a more heterogeneous training dataset that included both species present in Cabrera and additional taxa unique to the larger islands ([Table S5](#)). To confirm that site-pairs within each habitat acted as true ecological replicates, we calculated Bray–Curtis dissimilarities of plant species abundances (see [Figure S1](#)).

2.3 | Automated and manual interaction data acquisition

Automatic monitoring was conducted using three camera versions of the ACS device: ACS-V2, ACS-HQ and ACS-NoIR. This setup allowed us to investigate how different camera characteristics may influence pollinator detection. Diurnal monitoring employed ACS-V2 and ACS-HQ units recording video at 24 frames per second (fps), that is, one frame every 0.04 s. Although the cameras are capable of higher frame rates, 24 fps is sufficient to detect pollination visits involving contact with the reproductive parts of the flower—such as pollen deposition or collection—which typically occur over longer time spans. We scheduled video recordings of 10-min, followed by a 50-min pause, in three consecutive loops from 10:00 to 13:00 and again from 15:00 to 18:00. This looped schedule enabled repeated sampling of the same plant individuals while minimising temporal correlation between pollinator visits and also contributes to being both energy- and memory-efficient. ACS-V2 units were positioned 15 cm from the focal plants, while ACS-HQ units were placed at 40 cm, to account for differences in

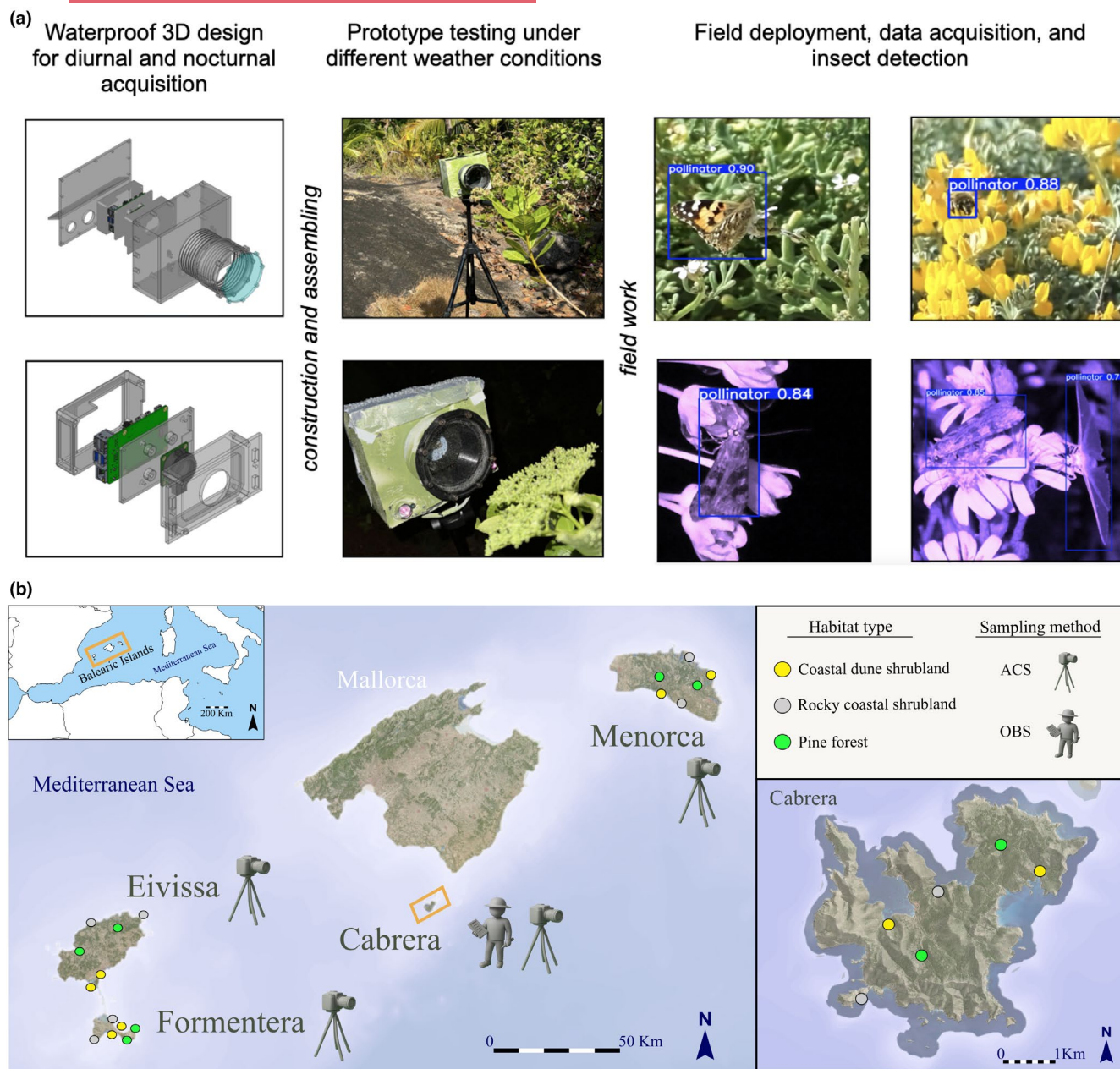


FIGURE 1 (a) Pipeline used for automatic plant–pollinator data acquisition, including video capture and pollinator detection. Left column: Design of the Automatic Camera System (ACS), the waterproof external case (top) and the inner case (bottom). Central column: Field deployment of the ACS during day (top) and night (bottom) period. Right Columns: Examples of two frames of a nocturnal video (bottom) and two from diurnal video (top) with some identified pollinators. (b) Map showing the location of study sites across different habitat types on each island, including the two replicates per habitat, where automated plant–pollinator data acquisition was conducted. Direct (manual) observations were carried out exclusively on Cabrera Island.

image resolution and focal length. The distance between the lens and the image sensor when the subject is in focus affects factors such as field of view. As a result, ACS-HQ units captured a focused area of approximately 10×8 cm, whereas ACS-V2 units covered a larger area of about 15×10 cm. Nocturnal monitoring was performed using ACS-NoIR units, which operated for 10-h periods with 10 loops per session at 15 cm from the plant, covering an area of about 15×10 cm. Deployment schedules were adjusted seasonally to account for changes in sunset and sunrise, ensuring optimal

coverage. In early spring (sunset $\approx 19:00$), ACS-NoIR operated from 19:00 to 05:00, whereas in June (sunset $\approx 21:00$), recordings were shifted to 21:00–07:00. ACS units were retrieved from the field after completing the observation periods. Two ACS units were deployed per flowering plant species, resulting in a total of 12 ACS units operating simultaneously at a site when six plant species were in bloom. To evaluate ACS performance in plant–pollinator interactions data collection for a community, we compare it to the data collected manually.

Direct observations were conducted in parallel with automatic recordings at the six study sites on Cabrera Island (in 2022 and 2023), across 13 field campaigns, each covering all sites (see Figure 1b). Two trained field technicians carried out 10-min observation windows between 10:00–14:00 and 14:30–18:30, aligning with the ACS diurnal recording schedule. This coordination ensured that both sampling methods were exposed to the same environmental conditions, thereby minimising the risk of differential effects on pollinator activity. To ensure temporal coverage across the day, the technicians rotated among plant species, observing a different species during consecutive censuses. For species with sufficient individuals, a total of 60 min of observation was conducted, while those with sporadic presence (one or two individuals) received 40 min of observation. To avoid observer interference and double-counting, individuals monitored by the ACS units were excluded from the direct observations. Each census consisted of monitoring a single plant individual for pollinator visits, enabling standardised and comparable data collection across sites.

Data acquisition in Cabrera was conducted under sunny conditions, with temperatures ranging from 13.2°C to 31°C. ACS-NoIR units were tested during calm nights, with humidity levels fluctuating between 23% and 91%, and occasional sporadic rainfall (data from the Spanish governmental meteorological agency, *Agencia Estatal de Meteorología* (Spanish State Meteorological Agency (AEMET), 2025)). In total, ACS units and field technicians were active for a total of 36 days, yielding approximately 153 h of automatically acquired data and 227 h of manually acquired data.

Both methods—ACS (based on video review) and direct observations—recorded standardised data, including *start* and *end* time, *plant species*, *total number of open flowers*, *number of monitored flowers*, *pollinator species* (or genus/family when species-level identification was not possible), *number of pollinator individuals* and the *number of contacted flowers*. We counted as interaction only those visits that contacted the reproductive part of the flower for at least 1 s. For ACS monitoring, all automatically detected pollinators were manually validated. Interactions that did not meet the minimum contact time—or involved individuals merely flying past without contacting any flower—were discarded. In the direct observations, any pollinators that could not be identified in the field were collected for later taxonomic identification by entomologists. In the case of ACS videos, species identification was conducted by an entomologist after detection. When identification was uncertain, a second entomologist reviewed the footage to confirm or reject the identification. In both methods, insects that could not be identified to the species, genera or family level were labelled as *Unknown Insect*.

Accounting for consecutive visits of the same individual is a challenging issue to address for both methods. For this reason, we ensured that a consistent protocol was followed. When a pollinator entered the field of view (ACS or direct observation) and visited multiple flowers in sequence without leaving the visual frame, we recorded this as one individual of a given species making several floral contacts. However, if the pollinator exited the field of view and later a conspecific individual appeared, we could

not ascertain whether it was the same insect or a different one. To ensure consistency and avoid underestimating the number of visitors, we adopted a conservative approach and systematically counted such reappearances as new individuals.

A summary of the key characteristics of both datasets from Cabrera Island is provided in Table 1. These are the datasets that will be used for the ecological analysis of both sampling methods, Sections 3.3 and 3.4. The data are aggregated across all field campaigns, sites and methods.

2.4 | Assembling the dataset for the pollinator detection algorithm

Building a high-quality dataset is fundamental for developing a robust deep learning model capable of accurately detecting pollinators in diverse ecological contexts (Stark et al., 2023). Our aim was to train a general pollinator detector, independent of fine-scale taxonomy, to ensure applicability across diverse natural monitoring scenarios. To this end, we compiled and annotated 7271 field-acquired images extracted from videos recorded by ACS units, capturing a variety of pollinators visiting a variety of plant species. Each frame obtained from ACS was manually labelled using Labellmg 1.4.0 (Tzutalin, 2015) and Roboflow (Dwyer et al., 2024), with precise bounding boxes (indicating the exact location of the pollinator in the image) and, when possible, identification at the species, family or functional group level. However, given that our primary goal was pollinator detection independent of taxonomic identity, we merged all species, family and functional group labels into a single 'pollinator' class, encompassing large bees, small bees, coleopterans, small coleopterans, dipterans, small dipterans, hoverflies, lepidopterans, wasps and other arthropods (including primarily ants and thysanopterans), moths and Unknown insects (see Table S1) alongside a diverse range of plant species and individuals (see Table S2). Vertebrate pollinators (i.e. birds and lizards) were excluded from model training due to their distinct morphological features. To further increase both the diversity and scale of our training corpus without additional fieldwork, we augmented this initial dataset with 4151 publicly available images—851 drawn from the ArTaxOr dataset (Mazen, 2023) and 3300 sourced from various open pollinator collections on Roboflow (Dwyer et al., 2024).

From the original set of acquired images, to further improve the model's robustness and generalisation, we applied data augmentation techniques (Shorten & Khoshgoftaar, 2019), ensuring the model could detect pollinators across varying environmental conditions and imaging perspectives. These augmentations techniques included the following. Horizontal flipping ($p=0.5$): randomly flipping half of the images to introduce variability in pollinator orientation. Translation (± 0.1): shifting images up to 10% in both horizontal and vertical directions to help the model recognise pollinators at different positions. Scale variations (± 0.5): adjusting pollinator sizes by $\pm 50\%$, ensuring detection at varying distances. Additionally, mosaic augmentation was performed, merging four different images into a single composite image, which enhanced

Plant species	C _{obs}	C _{acs}	H _{obs}	H _{acs}	F _{obs}	O _{obs}	F _{acs}	O _{acs}
<i>Cakile maritima</i>	179.0	32.0	31.5	21.5	31.1	33.4	19.4	53.5
<i>Teucrium capitatum</i>	164.0	19.0	28.3	12.3	33.1	36.2	35.9	78.3
<i>Daucus carota</i> L. subsp. <i>majoricus</i>	156.0	17.0	26.0	11.3	8.3	8.4	17.3	41.7
<i>Limonium</i> sp.	131.0	23.0	30.2	17.5	135.7	218.4	49.5	82.0
<i>Cistus monspeliensis</i>	128.0	24.0	22.2	16.7	47.6	64.8	22.7	52.8
<i>Sedum sediforme</i>	59.0	9.0	10.3	5.5	8.4	8.5	10.6	11.3
<i>Fumana ericoides</i>	56.0	5.0	13.0	4.2	3.7	3.7	4.0	4.0
<i>Helichrysum stoechas</i>	52.0	14.0	9.8	9.8	55.4	64.9	60.6	127.4
<i>Asteriscus aquaticus</i>	49.0	7.0	8.8	5.2	10.4	10.5	13.3	13.9
<i>Euphorbia dendroides</i>	49.0	5.0	8.3	3.2	97.4	214.8	51.4	600.0
<i>Astragalus balearicus</i>	38.0	15.0	6.3	11.8	122.7	122.7	215.2	310.2
<i>Globularia alypum</i>	35.0	4.0	8.0	3.3	2.5	2.5	1.2	1.2
<i>Asphodelus fistulosus</i>	24.0	2.0	4.0	1.7	8.5	8.9	9.0	12.0
<i>Allium</i> sp.	21.0	2.0	3.5	1.5	6.4	6.4	2.0	3.0
<i>Dorycnium fulgurans</i>	19.0	6.0	3.2	4.5	272.4	384.9	156.7	330.0
<i>Ruta chalepensis</i>	16.0	5.0	2.7	4.2	7.6	7.6	10.2	11.4
<i>Salvia rosmarinus</i>	15.0	11.0	2.5	8.5	97.5	117.6	56.6	94.8
<i>Euphorbia serrata</i>	14.0	7.0	2.3	5.8	7.7	7.7	14.6	17.6
<i>Blackstonia perfoliata</i>	11.0	2.0	2.7	1.3	5.6	5.6	6.0	6.0
<i>Lonicera implexa</i>	9.0	1.0	1.5	0.8	37.3	37.3	60.0	110.0
<i>Allium ampeloprasum</i>	7.0	2.0	1.2	2.3	1.0	1.0	1.0	1.0

Note: C_{obs} and C_{acs} represent the total number of plant individuals censused. H_{obs} and H_{acs} correspond to the monitoring hours. F_{obs} and F_{acs} indicates the mean number of monitored flowers per census. O_{obs} and O_{acs} are average amount of open flowers for plant species. Note that the average number of open flowers is generally higher than the number of monitored flowers, as both the camera and field technicians may only observe a fraction of the total open flowers, particularly when the plant individual (often a shrub) is too large or has too many flowers in bloom.

TABLE 1 Description of the dataset obtained through direct observations (obs) and the Automatic Camera System (acs) on Cabrera Island during the spring seasons of 2022 and 2023.

the model's ability to generalise across diverse backgrounds, plant structures and interaction contexts.

After field data collection, the dataset comprised the following entries: the diurnal ACS-V2 subset included 5878 images, while the diurnal ACS-HQ subset contained 1198 images. The nocturnal ACS-NoIR dataset consisted of 195 images. Across all subsets, there was an average of 1.1 pollinator annotations per image. When combined with the open-source images and data augmentation, the final dataset reached 21,435 images tagged, which were used to train multiple YOLOv5 model variants (see Section 2.5).

2.5 | Training and evaluating the pollinator detection model

To develop an accurate and efficient system for pollinator detection, we adopted the YOLOv5 architecture (Jocher et al., 2020), a state-of-the-art single-stage convolutional neural network that performs object localisation and classification in a single forward pass. This architecture is well suited for field-based ecological monitoring due to its balance of high detection accuracy and fast inference speed,

enabling real-time or large-scale video processing with limited computational resources.

We trained six YOLOv5 models using progressively broader subsets of data to evaluate performance under a range of imaging conditions. The first model was trained exclusively on the diurnal ACS-V2 dataset, while the second model used the higher-resolution ACS-HQ dataset to explore the effect of image quality. The third model focused on the nocturnal ACS-NoIR dataset, optimised for detection in low-light conditions. The fourth model was trained on the complete ACS dataset, integrating images from all three devices to assess generalisation across varied lighting environments. In the fifth model, we extended this dataset by merging the ACS images with open-source pollinator datasets to improve robustness and adaptability. Finally, the sixth model used the same combined dataset but incorporated SAHI (Slicing Aided Hyper Inference) (Svenning et al., 2025), a tiling-based method designed to improve the detection of small or partially occluded pollinators in high-resolution images.

All models were trained for 150 epochs with an image size of 1024×1024 pixels, using stochastic gradient descent (SGD) optimiser with an initial learning rate of 0.01, momentum of 0.937 and weight decay of 0.0005. The training schedule included three

warm-up epochs with momentum of 0.8 and bias learning rate of 0.1. Object detection parameters were configured with box loss weight of 0.05, classification loss weight of 0.5, objectness loss weight of 1.0 and IoU threshold of 0.2. Training ran on an NVIDIA RTX 3080 Ti GPU with 12GB GDDR6X memory, 10,240 CUDA cores, 1.67 GHz boost clock and 912 GB/s memory bandwidth, ensuring efficient neural network computation.

To assess performance, the dataset was divided into training, validation and test sets, with the specific split fractions varying depending on the experiment. In general, the training set ranged from a minimum of 60% to a maximum of 80%, while the test set ranged from 10% to 20%. In the specific case of the nocturnal image dataset, the training, validation and test sets were stratified to ensure that plant species represented during training were also present during evaluation, given the limited size of the dataset. However, we ensured that no image used for training was included in the validation set, thereby preserving the integrity of the evaluation. Five-fold cross-validation was performed in all experiments. Finally, to test generalisation to unseen flora, we ran a leave-one-species-out cross-validation on the ACS-HQ set (Sreedharan et al., 2023), removing all images of one plant species per split for validation. In all experiments, the test set, consisting of independent data not seen during training, provided an unbiased evaluation of model generalisation.

Each model was evaluated in terms of four key, well-known metrics: (1) Precision represents the proportion of correctly identified pollinators (true positives) out of all detections made by the model (true positives and false positives). A high precision indicates that the model is reliable in identifying the pollinators. That is, when it draws an identification, it is likely to be correct. (2) Recall (also known as sensitivity in some areas) measures the model's ability to identify all relevant pollinators present in the images. It is obtained as the ratio of true positives to the sum of true positives and false negatives. A higher recall means the model identifies most pollinators in the images, minimising the amount of false negatives. (3) mAP50 (mean Average Precision at IoU 0.5) evaluates the detection accuracy by computing the average precision for each class when the predicted bounding boxes align with the ground-truth bounding boxes by at least 50%. (4) The F1-score is the harmonic mean of precision and recall, offering a single measure that balances both metrics. It is particularly useful when seeking a balance between the model's ability to avoid false positives (precision) and false negatives (recall).

This structured training approach facilitates a direct comparison of model performance across different imaging conditions, enabling the selection of the most effective pollinator detection pipeline for field deployment.

2.6 | Diversity indices and plant–pollinator network metrics

To evaluate differences between plant–pollinator data collected through direct observations and ACS monitoring, we analysed key ecological patterns across multiple analytical scales using widely

applied diversity indices and network metrics (Dormann et al., 2014; Oksanen et al., 2007). We assessed diversity using two key parameters: *species richness*, representing the total number of unique pollinator species detected, and *Shannon diversity*, which incorporates both species richness and the evenness of species abundances. In addition, we quantified three types of pollination activity: (i) pollinator visitation rates, defined as the number of visits per pollinator species; (ii) plant visitation rates, indicating the number of visits received by each plant species, adjusted for flower availability; and (iii) total flower visits per pollinator individual, to capture potential behavioural differences. To enable fair comparisons between ACS and direct observation data, all visitation rates were normalised by both flower availability and the observation time specific to each method.

Quantitative bipartite networks were constructed using plant visitation rates as interaction weights. Network metrics included *Specialisation degree* for both plants and pollinators, which ranges from 0 (generalist) to 1 (specialist). Low values indicate pollinators with broad resource use whereas high values imply highly selective pollinators with exclusive partners (Dormann et al., 2008). We also calculated four network-level metrics to evaluate interaction structure, specialisation and stability, following the approach of Dormann et al. (2008). These metrics provide insights into the organisation of pollination networks and the extent to which each monitoring method captures ecological complexity. *Connectance*, which measures the proportion of realised interactions relative to all potential interactions, indicates how densely connected a network is. However, it does not account for the distribution of interactions among species. *Interaction evenness* assesses the distribution of interactions, revealing whether pollination activity is dominated by a few species or more evenly spread. *Interaction evenness* ranges from 0 to 1. Low values (close to 0) indicate a highly uneven or skewed distribution of interactions, where one or a few species pairs account for the majority of interactions, while most others occur rarely. High values (close to 1) reflect a more balanced distribution, where no particular interaction dominates and all occur with similar frequency. Ecologically, high interaction evenness suggests a generalist or highly uniform network, in which all interactions contribute more equally and there is no strong dependence on any single species pair. *Network specialisation* quantifies the degree of specialisation at the community level, ranging from 0 (fully generalist) to 1 (fully specialised), indicating how structured the network is in terms of pollinator preferences. *Robustness* evaluates the resilience of the network to species loss, measuring its ability to maintain structure and function as species are progressively removed. This metric is widely used to assess the stability and persistence of ecological networks and, as in the case of specialisation, it ranges from 0 (total collapse) to 1 (completely resilient).

To assess differences in diversity and network structure between ACS and direct observations, we compared *species richness* and *Shannon diversity*. We also analysed four key network metrics: *Specialisation index* at species level for both plants and pollinators, *Connectance*, *Interaction evenness*, *Network-level specialisation* and *Robustness*. To test if there were significant differences within the

index and metrics mentioned between methods (direct observations and ACS), we performed the paired test: Wilcoxon signed-rank test. To further evaluate how each method captures pollination networks, we analysed the interactions detected exclusively by one method but missed by the other (Bascompte et al., 2003; Memmott et al., 2004). Identifying these missed interactions offers insights into potential biases and complementary strengths of automated versus manual monitoring approaches.

All analyses were conducted using Python 3.12.1 (Python Software Foundation, 2023), leveraging the `pandas` (Reback et al., 2020), `seaborn` (Waskom, 2021) and `matplotlib` (Hunter, 2007) libraries for data processing and visualisation. Diversity indices and network metrics were calculated separately in R (R Core Team, 2025) using the `vegan` (Oksanen et al., 2007) and `bipartite` (Dormann et al., 2014) packages.

3 | RESULTS

3.1 | Field deployment evaluation

During field deployment, the ACS casings exhibited only minor surface wear, while all Raspberry Pi 4 hardware remained fully functional, with no recorded failures due to overheating or high humidity. To further evaluate the durability of the outer casing, we tested different 3D printing materials and coatings under field conditions. We observed that PLA filaments used for the outer casing experienced some deformation when printed in black and exposed to direct sunlight for extended periods. This is likely due to PLA's glass transition temperature of 55–60°C, which can be exceeded under high solar radiation. However, applying colour paint coatings significantly reduced deformations, suggesting that surface treatments can enhance material stability. In contrast, inner casings showed no signs of deformation due to Raspberry Pi heat, and no UV-related material deterioration was detected in any components. Based on these findings, we recommend revising the choice of 3D printing filament and material colour for future field campaigns. Specifically, using heat-resistant filaments and lighter-coloured materials could further improve casing durability in high-temperature environments.

3.2 | Pollinator detection algorithm evaluation

Results evaluating the performance of the YOLOv5 model for pollinator detection across different datasets is summarised in Table 2. The reported results represent the average performance across the five folds. Overall, the analysis shows consistent and satisfactory accuracy trends across variations in camera type, dataset size and type and imaging conditions.

The diurnal model trained on ACS-V2 camera data exhibited balanced performance, with an average F1-score of ≈ 0.86 . These results indicate that the model effectively detects pollinator interactions under daylight conditions, balancing precision and recall. The diurnal model trained on ACS-HQ data outperformed ACS-V2, achieving a precision of 0.93, a recall of 0.91 and an mAP50 of 0.95. Despite being trained on a smaller dataset, its higher-quality images likely improved detection accuracy, particularly for smaller pollinators. These results highlight the importance of image resolution in enhancing model performance. The nocturnal ACS-NoIR model, despite being trained on a very limited dataset, achieved a good precision (0.98). However, its lower recall (0.68) and F1-score (0.80) suggest that while the model is highly accurate when detecting a pollinator interaction, it misses a proportion of them, likely due to lower contrast and reduced visibility in nighttime conditions. When testing a model trained with the full dataset, that is combining all available data from both nocturnal and diurnal cameras, the performance metrics slightly improve while maintaining a good balance between precision and recall, reaching a final F1-score of 0.95. To complement our experimental analysis of pollinator detection, we also tested several variations of the initial dataset and models to explore whether model performance could be further improved. First, as detailed in the Methods section (Section 2.5), we experimented with including external datasets to increase image variability and improve model robustness. As shown in Table 2 (row 5), there is no gain in model performance compared to training only the detections captured within our own dataset. This suggests that our initial training dataset is sufficient for detecting pollinators in the selected habitats. The second experiment aimed to assess whether detection of small pollinators could be improved, given the expected limitations in this area (Tariq & Javed, 2025). In this case,

TABLE 2 Performance of pollinator detection by four different YOLOv5 models on videos captured with the Automatic Camera System (ACS) equipped with an Autofocus-Camera Module 2 (ACS-V2), a High-Quality Camera (ACS-HQ) or infrared-sensor cameras (ACS-NoIR).

Active period	Model	Train/val/test	Precision	Recall	mAP50	F1-score
Diurnal	ACS-V2	4877/345/656	0.87	0.85	0.85	0.86
Diurnal	ACS-HQ	862/92/244	0.93	0.91	0.95	0.92
Nocturnal	ACS-NoIR	158/18/19	0.98	0.68	0.81	0.80
Diurnal & nocturnal	ACS data Merged	10,192/3542/3550	0.95	0.95	0.97	0.95
Diurnal & nocturnal	ACS data + & Ex. Database	12,682/4373/4380	0.93	0.93	0.95	0.93
Diurnal & nocturnal	ACS data & Ex. Database + SAHI	12,682/4373/4380	0.80	0.90	0.85	0.85

Note: Ex. Database corresponds to the ACS supplemented with an external pollinator image database. SAHI (Slicing Aided Hyper Inference) is an inference approach to improve detection of small and partially occluded insects. Each row reports the active period of the day, the dataset used, the sizes of the dataset (training, validation and test) and the corresponding performance metrics (precision, recall, mAP50, F1-score).

the use of specialised tools such as SAHI (Akyon et al., 2022) (with a window size of 640 pixels), which are designed to enhance small object detection, did not lead to improved performance—in fact, it resulted in a slight decline. This suggests that, on one hand, such tools—typically applied to large-scale images—may not effectively address the challenges posed by our specific setup. On the other hand, it may indicate that the core difficulty may lie in the combination of small object size and complex, heterogeneous backgrounds, which remain challenging for current detection models. See results in row 6 of Table 2.

Finally, to evaluate how the model performs on previously unseen plant species, we conducted leave-one-out experiments at the plant species level. The results are presented in Table S3. As observed, the models tend to perform poorly in these setups, despite achieving good results on the validation set. This highlights the challenges associated with generalising to novel backgrounds not seen during training.

To complement the quantitative evaluation of deep learning metrics, we conducted a manual review of diurnal and nocturnal video samples to qualitatively assess model performance (see Figure 2). This assessment aimed to identify potential sources of detection errors and evaluate how sensor quality influenced pollinator

classification accuracy. Across 12 manually reviewed videos, the model correctly identified 43 true positives, while 7 false positives and 8 false negatives were recorded (those results are provided in Table S4). False negatives primarily involved small dipterans and some bee species, which the model failed to detect, whereas false positives were mostly misclassified leaves, incorrectly identified as pollinators. Performance varied among ACS units due to differences in sensor resolution and optical quality. The ACS-V2, capturing 5-megapixel images, often struggled to detect small-bodied pollinators, leading to higher false negatives. In contrast, ACS-HQ, with its higher-resolution video output, demonstrated improved detection accuracy, effectively distinguishing both small pollinators and larger functional groups. Additionally, the ACS-NoIR system, designed for nocturnal monitoring, benefited from high-contrast imagery against dark backgrounds, which enhanced the detection and labelling of moths visiting flowers at night.

3.3 | Ecological indices and visitation patterns

Direct observations covered a total of 56,789 flowers and recorded 3273 pollinator individuals, which contacted 6483 flowers over



FIGURE 2 Detection performed through YOLOv5 trained algorithm. The panels show: (a) four true positives (*Bombus hispanica*, two *Microdiptera* sp. and a *Colletes* sp.); (b) one false positive, both on *Cakile maritima*; (c) one false negative; and (d) one true positive (*Anthophora* sp.) from similar frames of the same video on *Lotus cytisoides*.

Metric	OBS (mean $\pm$ SD)	ACS (mean $\pm$ SD)	p-value
Diversity indices			
Species richness	7.8 $\pm$ 4.10	7.00 $\pm$ 4.16	0.34
Shannon diversity	1.49 $\pm$ 0.54	1.49 $\pm$ 0.63	0.76
Network-level metrics			
Plant species specialisation	0.20 $\pm$ 0.12	0.21 $\pm$ 0.11	0.44
Pollinator species specialisation	0.28 $\pm$ 0.16	0.28 $\pm$ 0.14	0.77
Connectance	0.55 $\pm$ 0.17	0.67 $\pm$ 0.25	0.02*
Network-level specialisation	0.26 $\pm$ 0.07	0.30 $\pm$ 0.07	0.01*
Interaction evenness	0.58 $\pm$ 0.08	0.58 $\pm$ 0.05	0.86
Robustness	0.50 $\pm$ 0.08	0.44 $\pm$ 0.07	0.02*

TABLE 3 Summary table of the two monitoring methods: Direct Observation (OBS) and Automatic Camera Systems (ACS), based on data from 13 field campaigns across 6 study sites.

Note: For each metric, the table presents the mean $\pm$ standard deviation and the corresponding p-value, with asterisks indicating statistically significant differences. Statistical comparisons were performed using the Wilcoxon signed-rank test (paired Wilcoxon test).

228 h. ACS monitoring, in contrast, observed 9468 flowers and detected 1624 individuals performing 2862 pollination events across 153 h. Despite these differences in sample size and effort, *species richness*, *Shannon Index* and specialisation degree at the species level for both plants and pollinators did not differ significantly between methods (see Table 3).

To assess the reliability of both methods in capturing the full diversity of pollinators, it is essential to evaluate and compare the completeness of the data collected, as this may influence the interpretation of the diversity metrics reported. Since no external database exists for Cabrera Island that documents all pollinator species, we analysed how cumulative species richness evolves with increasing sampling effort. The results, presented in Figure S2, show that four out of the six sites appear to approach a saturation point, suggesting that a substantial portion of local biodiversity has been captured. In contrast, the species accumulation curves for the two remaining sites continue to rise, suggesting incomplete coverage. The plots also provide a valuable opportunity to compare the two sampling methods. Overall, both methods exhibit similar trends; however, in two specific sites, one Rocky Coastal and one Dune System, one method appears to underperform relative to the other in terms of species coverage. This discrepancy may be due to the selection of specific plant species at those sites, which could have influenced the diversity observed with the method. Finally, we find that combining both sampling methods is advantageous, as the combined species richness significantly exceeds that of either method used independently.

From the pollinator point of view, a comparison of the recorded frequency of visits between the two methods reveals that ACS recorded, in general, a higher frequency of interactions. This observation is illustrated in the main panel of Figure 3 and the accompanying inset. However, some pollinator species were exclusively detected through direct observations, particularly small-bodied insects (body length <5 mm), such as the dipterans *Acanthiophilus helianthi*, *Agromyzidae* sp. and *Sphenella marginata*, as well as the wasps *Gasteruption* sp. and *Tachysphex obscuripennis*. These results suggest that ACS may struggle to detect insects below a certain size

threshold due to image quality constraints. Additionally, five pollinator species larger than 5 mm were also recorded exclusively through direct observations, including *Lydella* sp., *Muscidae* sp., *Thyridanthrax elegans*, *Stomorphina lunata* and the hemipteran *Centrocoris* sp. All these undetected species were smaller than 5 mm. In contrast, ACS uniquely recorded interactions involving *Bombyliidae* sp. and the bird *Phoenicurus ochruros*. It is also interesting to analyse the last column of the figure, which corresponds to insects that could not be assigned to any species.

Turning now the attention to plants, Figure 4a compares the (normalised) number of pollinator visits per plant species disregarding pollinator species. Direct observations recorded a greater number of pollinators in comparison to ACS monitoring, across most plant species. The difference was most pronounced in high-activity pollination environments, especially for *Dorycnium fulgurans* and *Salvia rosmarinus*. At first glance, this observation may suggest that automated systems have technical limitations in accurately capturing some pollinator visits; however, a detailed analysis reveals an ecological explanation, facilitated by the lizard *Podarcis lilfordi*. Increased lizard activity may increase the pollination rate for this plant species, but it could also discourage insect pollinators due to its predatory behaviour, thereby introducing a dual effect on the results presented in Figure 4. In *Euphorbia dendroides*, higher lizard pollination activity has been detected in the ACS dataset, suggesting that direct monitoring may have underestimated its role in the ecosystem, likely due to the presence of field observers. This, in turn, may have reduced, in the ACS recordings, the high frequency rates of some pollinators visiting several plant species such as *Salvia rosmarinus*, *Limonium* sp. or *Euphorbia serrata* recorded by direct monitoring. We observe significant differences in visit frequencies (Figure 4), as well as increased interaction frequencies between *P. lilfordi* and the aforementioned plant species in the ACS dataset (see Figure 5). These findings evidence that vertebrate pollinators can influence pollination networks by simultaneously facilitating and displacing insect pollinators, which may affect method-specific detection biases.

Regarding pollinator behaviour, across practically all plant species (see Figure 4b), ACS recorded a similar number of per-capita

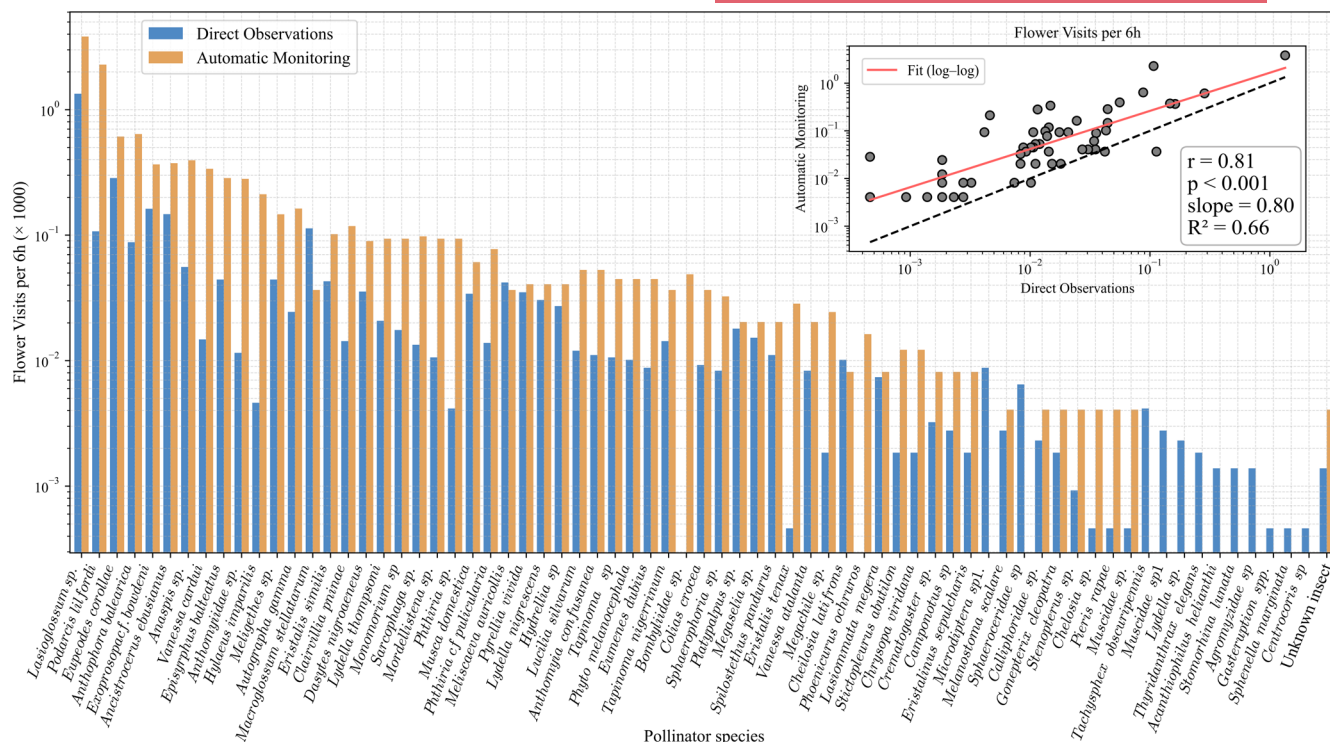


FIGURE 3 Average pollinator visits per pollinator species by two methods: Direct field observations (blue) and an Automatic monitoring (orange). The values are calculated by dividing the total number of recorded pollinators by the overall census time and the number of monitored flowers. Species are ordered by the combined visit rate (highest to lowest). The last column represents 'Unknown insect', referring to individuals that could not be classified to the species level due to uncertainty or lack of clear visual information. Inset: Log-log scatter of per-species visit rates, showing the 1:1 reference line (black dashed) and the log-log regression fit (red). Statistics from the regression show Pearson's correlation coefficient (r), p -value, slope and R^2 .

pollinator visitation probability compared to direct observations. Only *Cistus monspeliensis* and *Euphorbia dendroides* showed elevated values, largely driven by *P. lilfordi*, which recorded higher flower visitation rates in ACS censuses (30 and 106, respectively) than in direct observations (17 and 58, respectively).

3.4 | Plant-pollination interaction network

In this section, we take a deeper look at the datasets obtained from ACS monitoring and direct observation from the perspective of plant-pollinator networks. Returning to the results provided in Table 3, we observe that several network metrics reveal significant differences. *Connectance* was higher in ACS-derived networks, suggesting that automated monitoring captured a greater proportion of potential interactions within the system. Similarly, *network specialisation* was slightly but significantly higher in ACS, as low-frequency events are easier to detect with ACS monitoring. Note that the probability of detecting a pollinator that visits a flower once per hour during a 10-min census is only 17%, highlighting the limitations of short observation periods in direct monitoring. In contrast, *robustness* was greater in networks derived from direct observations, likely because low-frequency interactions detected by ACS were more transient and prone to disappearance. Despite these differences, both

datasets exhibited comparable interaction evenness, with a similar distribution of interaction frequencies. To complement these results, Table S5 presents this analysis disaggregated by habitat type and year for those metrics that presented significant differences in Table 3. Regarding habitat type, the most significant differences are observed in the rocky coastal habitats. We also observe variation when analysing each year independently, with statistically significant differences primarily appearing in the 2023 data. Based on these results, we conclude that both methods may exhibit differences that require specialised analysis beyond what aggregated statistics can reveal. Despite this, we can confirm that, overall, both methods capture different information with respect to these network indices.

To further explore discrepancies between networks generated by both methods, we examined the interactions detected by one method but not with the other in Figure 5. Overall, direct observations recorded fewer missing interactions ($n=65$) compared to ACS monitoring ($n=131$), where a missing interaction is defined as one observed using a single monitoring method but absent in the other. Taking a closer look at individual differences, we can clearly detect many variations in several plant-pollinator relationships, such as those involving *Teucrium capitatum* and *Vanessa cardui*, as well as others mentioned in the previous section. However, the most evident difference is the distinct role played the lizard *Podarcis lilfordi*, particularly in its interactions with *Astragalus balearicus*, *Cistus*

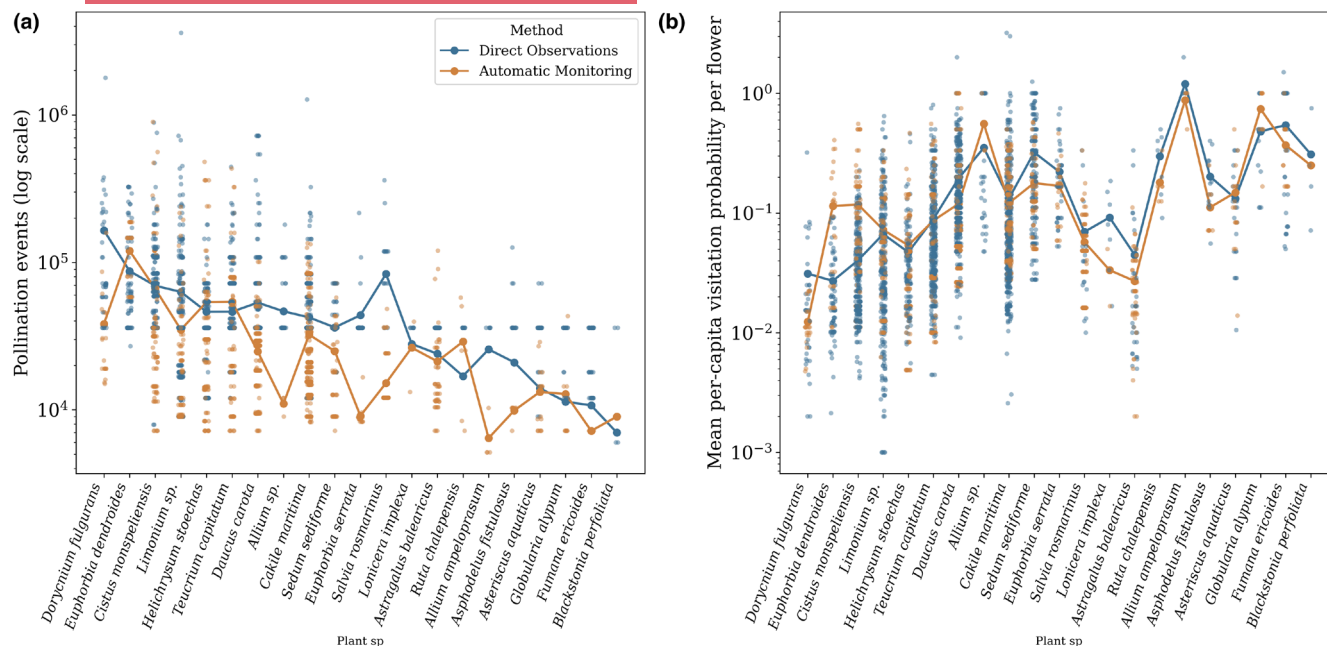


FIGURE 4 (a) Average pollinator visit rate per plant species over a standardised six-hour period, shown separately for each monitoring method. Visit rates were calculated by dividing the number of pollinator observations by census duration and the number of flowers surveyed, then scaling to the average number of open flowers per species and extrapolating to a six-hour observation window. Each point corresponds to the individual values per census, with lines indicating averages. The plot is ordered in decreasing order based on the average number of events, considering both direct observations and ACS detections. (b) Mean per-capita visitation probability per flower, defined as the number of observed visits to flowers divided by the number of observed pollinators and the number of monitored flowers. Note that the denominator represents the total number of potential interactions between the observed pollinators and the monitored flowers, while the numerator reflects the actual number of detected visitation events.

monspeliensis, *Euphorbia dendroides*, *E. serrata*, *Helichrysum stoechas*, *Linonium sp.* and *Salvia rosmarinus*, where both the number of individuals and flowers visited were much higher in ACS records than in direct observations. This increased presence may have influenced the counts of other pollinators and could explain the lower number of insect pollination events recorded in other plants with which the lizard interacts, given its role as a predator. For instance, ACS recorded 57 visits by *M. auricollis* to *E. dendroides* out of the 81 interactions detected through direct observations, likely influenced by high *P. lilfordi* activity. These findings provide evidence that human presence during direct observations alters pollinator behaviour, particularly in plants frequently visited by vertebrate pollinators, indirectly affecting insect visits.

To conclude our analysis, with the support of Figure 6, we attempt to explain why each of the two methods fails to detect certain interactions. Figure 6a represents the frequency of each plant-pollinator interaction detected by ACS (x-axis, log scale) and direct observations (left y-axis). Each point represents a single interaction, with the right y-axis showing the number of overlapping missing interactions in direct observations at the same x-coordinate. The dashed trend line suggests that as ACS-detected interaction frequency increases, the likelihood of that interaction being missed by direct observations decreases. In other words, direct observations effectively capture frequent interactions but often miss sporadic ones, a limitation that ACS monitoring overcomes.

Figure 6b highlights the pollinator species that ACS monitoring has failed to detect. Most undetected interactions involved pollinators smaller than 5 mm, such as *Acanthiophilus helianthi*, *Sphenella marginata* and *Tachysphex obscuripennis*. In contrast, pollinators larger than 5 mm were consistently detected by both methods, suggesting that ACS performs well in capturing interactions involving larger pollinators, while its sensitivity to small-bodied insects may require further refinement, except for *Meliscaeva auricollis*, represented by the larger red point (approximately 8 mm in size), whose particular case had been explained previously. Additionally, in the panel, we observe a potential misleading effect for pollinators smaller than 10 mm, where the number of missing observations appears to decrease along the y-axis as the number of direct monitoring counts decreases. This pattern arises because rare pollinator events are less likely to be detected by direct observations, which in turn results in fewer missing interactions recorded by ACS monitoring. This situation is particularly challenging, as neither direct observation nor ACS monitoring can effectively capture low-frequency interactions of small individuals, at least with the current protocols used here.

4 | DISCUSSION

The integration of big data approaches in ecology increasingly depends on automated monitoring systems to enhance data collection



Combining Raspberry Pi units with YOLOv5 holds strong potential for automating pollinator monitoring in natural environments at the community level. The algorithm achieved good accuracy metrics across different model versions and configurations, including the use of additional tools. These promising results suggest that, in the future, it may be feasible to automatically monitor plant-pollinator

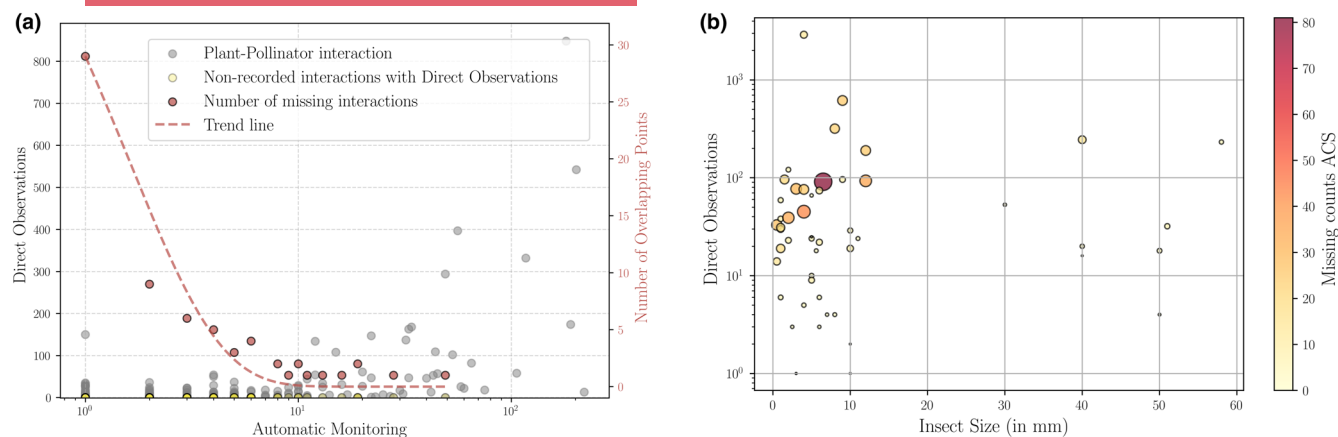


FIGURE 6 (a) Relationship between interaction frequencies recorded by automatic monitoring (x-axis, log scale) and direct observations (y-axis). Each grey point represents a unique plant–pollinator interaction detected by both methods. Yellow points at $y=0$ correspond to interactions detected exclusively by automatic monitoring. The red line (right y-axis) indicates, as well, the number of interactions not detected by direct observation and correspond as well to the number of yellow points that overlap at each x-value. The red line represents a trend line fitted to the red points, highlighting the decreasing tendency to miss detections as the interaction frequency increases. (b) Scatter plot where each point represents a pollinator species. The x-axis shows body size (in millimetres), and the y-axis indicates the number of pollination events recorded through direct observations (log scale). Circle size and colour denote the number of events detected by direct observations but not by the ACS monitoring. Taxa with zero detections in the direct observation dataset have been excluded.

interactions at the species level and occasionally classify rarer or less distinguishable insects to genus or family level.

Our comparative analysis across datasets revealed that image quality—rather than dataset size—is the primary factor influencing detection accuracy. Additionally, leave-one-out experiments indicate that the models struggle to handle background heterogeneity. This suggests that including a diverse range of plant species may improve model generalisation by exposing it to a broader variety of visual contexts. In summary, key factors affecting detection performance include the size of the insect within the frame, background complexity and depth of field, with narrower depth helping to distinguish pollinators from their surroundings. While YOLOv5 performed well in detecting pollinators across a range of body sizes, manual verification remains necessary for small-bodied insects, especially under suboptimal image quality conditions.

Our results indicate that deploying a sufficient number of ACS units to achieve optimal sampling completeness can yield estimates of species richness and abundance comparable to those obtained through direct observations. However, certain network metrics may still differ between methods due to variations in sampling strategy. While ACS units typically monitor fewer flowers than a human observer can cover during a direct observation, they offer a key advantage: the ability to remain in the field for extended periods—especially when solar-powered. This extended deployment allows ACS to capture different, yet equally valuable, aspects of pollinator activity. As such, ACS and direct observations should be viewed as complementary methods, each helping to balance the limitations and biases of the other. At present, however, video recordings must still be reviewed manually to extract plant–pollinator interactions at the species level. In this study, we therefore used ACS as a semi-automated monitoring tool, manually reviewing all footage to ensure

that interactions were accurately captured and that no relevant information was lost.

A distinctive factor influencing pollinator activity in Cabrera is the lizard *Podarcis lilfordi*, which not only feeds on insects but also functions as a pollinator. This species is highly abundant on the island and frequently visits flowers. Lizard pollination is a well-documented phenomenon on islands (Olesen & Valido, 2003), and the pollination effectiveness of *P. lilfordi* has been previously demonstrated (Traveset & Sáez, 1997). In Cabrera—where lizard densities are particularly high—their influence on plant–pollinator interactions may be stronger than in other systems, especially when compared to continental areas, though it may be comparable to that observed on other islands. Our results also suggest that human presence can affect vertebrate pollination. Specifically, lizards tend to contact fewer flowers when humans are nearby, which may reduce their pollination activity and indirectly influence insect visitation by removing a natural predator. In this context, ACS units offer a valuable tool for recording interactions under undisturbed conditions, providing a more accurate representation of natural pollination dynamics than direct human observation.

Our study highlights a key limitation of the ACS that may need to be addressed to improve its effectiveness. In species-rich plant communities, achieving adequate sampling may require deploying more ACS units or extending sampling over multiple days to ensure sufficient flower coverage, observation time and interaction counts per plant species. Another critical constraint is image resolution, which can hinder the detection and identification of small-bodied insects—a particularly important issue in island ecosystems, where small pollinators often play important ecological roles. The underdetection of groups such as small Diptera and solitary bees may bias estimates of pollinator diversity, interaction frequency and network structure. In

our study, pollinators smaller than 5 mm represented approximately 17% of the total diversity—modest in comparison to ecosystems dominated by small flowers, where these insects often prevail in richness and abundance (Agosta & Janzen, 2005). While larger pollinators tend to carry more pollen (Cortés-Gómez et al., 2023), size matching between pollinators and flowers is a known predictor of interaction frequency and pollination effectiveness (Naghiloo et al., 2021). Thus, in systems where small-bodied pollinators represent a significant portion of the community, their underrepresentation in automated datasets like those generated by ACS must be carefully considered, particularly in conservation or restoration contexts where they may fulfil rare or functionally critical roles. Addressing this challenge requires a twofold technical strategy: using the ACS-HQ system positioned closer to focal plants to improve resolution, and advancing deep learning-based detection models, such as YOLO algorithms (Ji et al., 2024), to better identify small insects against complex backgrounds, where they are often misclassified or missed entirely. To address the current limitations of automatic monitoring systems, future research should focus on automating classification directly from field data at broader taxonomic scales, such as functional groups or family levels. This approach is crucial, as accurate species-level identification requires extensive taxonomic expertise and large, collaboratively curated datasets (Bjerge, Alison, et al., 2023; Bjerge, Geissmann, et al., 2023). On the other hand, the study of plant–pollinator communities presents additional challenges. Insect detection and classification alone are not sufficient, as the task requires identifying actual pollination events. This involves verifying that pollinators come into contact with the reproductive parts of the flower, something that remains difficult to assess using fully automated methods. While future developments in deep learning algorithms may enable more reliable detection of such interactions, manual curation of the datasets is still necessary at present to ensure accuracy.

A critical question that remains is how well the presented ACS system would perform in other plant communities beyond those studied in the Balearic Islands. While our tool has proven successful in these specific Mediterranean habitats, its broader applicability depends on several factors. The hardware, housed within the 3D-printed case, should function similarly in other environments. However, the recording duration must be adjusted to match the expected frequency of pollinator interactions, and the daily sampling schedule should be adapted to local light conditions and regional photoperiods. The algorithms have been trained on a diverse dataset representative of Mediterranean plant communities. Nevertheless, when applied to regions or plant species with very different floral architectures, it may require fine-tuning. This can be done by recording additional videos with the ACS system to adapt the model to the specific visual and ecological context. Fortunately, since many insect pollinators share similar morphological features, the core detection model is expected to generalise relatively well. As we envision automatic monitoring tools for entire plant–pollinator communities, future versions of the algorithm should be able to detect a wide range of pollinators—from hummingbirds and large bees to micro-coleopterans.

Beyond methodological advancements, ACS offers a powerful tool for addressing broader ecological questions, including pollinator behaviour, interspecific interactions and competition. Its capacity for long-term video storage and repeated review provides valuable opportunities to study plant–pollinator networks at unprecedented temporal and spatial resolutions.

ACS also shows strong potential for expanding plant–pollinator community studies to include nocturnal data—an often overlooked but equally important component of pollination ecology. Nocturnal interactions have traditionally been underrepresented due to the logistical challenges of nighttime fieldwork, such as the need to remain in the field overnight using red-light illumination, and the limited accuracy of manual observations in low-light conditions. In this context, ACS helps overcome these constraints, offering an effective means of reducing bias in environments where nocturnal pollination has received little attention.

Our findings suggest that ACS is particularly well suited to complement manual surveys in remote or hard-to-access locations, such as isolated islands, high-altitude zones or high-latitude ecosystems. Deploying multiple ACS units enables researchers to increase both spatial and temporal resolution, while generating extensive video datasets for detailed analysis and future reanalysis. Moreover, the adaptability of ACS across different environments extends its applications beyond pollination studies, including early detection of invasive species (Lake et al., 2022) and pollination monitoring in agroecosystems (Singh & Joshua, 2024). The open-source nature of ACS fosters collaborative development and continuous refinement of detection algorithms, allowing models to be tailored to a wide range of ecological contexts. Ultimately, integrating automated systems like ACS with field-based methods enhances the depth and quality of ecological datasets, supporting more robust biodiversity conservation and ecosystem research.

In short, our study demonstrates that ACS systems can effectively monitor entire plant communities over extended periods, including challenging nocturnal interactions. Comparative analyses of diurnal ACS data and direct observations confirmed the reliability of ACS in constructing accurate plant–pollinator networks. By integrating these complementary datasets, ACS enhances data resolution and completeness, accounting for biases inherent to each monitoring approach. Our findings also highlight the potential of ACS to standardise automated acquisition of plant–pollinator interactions—both diurnal and nocturnal—while reducing reliance on continuous human presence. Capturing a wider range of interaction frequencies, including rare events and vertebrate pollination activities affected by observers, ACS significantly improves the resolution of plant–pollinator networks. As a flexible, scalable and open-source tool, ACS establishes a robust foundation for further methodological refinements and broader ecological and conservation applications.

AUTHOR CONTRIBUTIONS

Pau Enric Serra-Marin, Anna Traveset and Arancha Lana conceived the ideas. Pau Enric Serra-Marin, Anna Traveset and Sandra Hervías-Parejo designed the field methodology and collected the data;

Arancha Lana developed the deep learning algorithms, Albert Solé-Ribalta designed and printed the 3D casings; Pau Enric Serra-Marin, Albert Solé-Ribalta and Javier Borge-Holthoefer analysed the data; Pau Enric Serra-Marin and Albert Solé-Ribalta led the writing of the manuscript, and all authors contributed critically to the drafts and gave final approval for publication.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interests.

PEER REVIEW

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DATA AVAILABILITY STATEMENT

The plant–pollinator interaction dataset, collected through direct observations and the Automatic Camera System (ACS), together with the accompanying scripts, has been uploaded as files accessible to reviewers and editors. The plant–pollinator interaction dataset is publicly available at Serra Marin (2025), <https://doi.org/10.5281/zenodo.17130777>. The YOLOv5 algorithm is available at Lana (2025), <https://doi.org/10.5281/zenodo.17130918>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1. Pollinator functional groups, included in the YOLOv5 training dataset, indicating the number of individuals, and the representative species in each group.

Table S2. Plant species included in the YOLOv5 training dataset, indicating the number of individuals, data source, island of collection, and year of record. The ACS refers to the Automatic Camera System, with three configurations: V2 (wide-angle lens), HQ (high-quality lens with 35mm focal length), and NoIR (wide nocturnal lens equipped with infrared sensors).

Table S3. Performance metrics for the leave-one-out experiments. Each row corresponds to a test in which the indicated plant species was included only in the test set—meaning no examples from that species were used during training or validation. P refers to *Precision* and R to *Recall*. The *Validation* columns report results on the validation set, while the *Test* columns show the outcomes obtained during the test stage.

Table S4. Comparison of pollinator detections by YOLOv5 and manual review across diurnal and nocturnal video samples (Videos 1 to 12). Detection counts are shown as YOLOv5/Manual Review. The number of false positives for each video is reported in the bottom row. Videos were recorded using the Automatic Camera System, with three configurations: V2 (wide-angle lens), HQ (high-quality lens with 35mm focal length), and Noct (wide nocturnal lens equipped with infrared sensors).

Table S5. Results of Wilcoxon signed-rank (Paired Wilcoxon test) tests comparing methods for the Connectance, Network-level Specialisation and Robustness metrics by habitat and year.

Figure S1. Bray–Curtis dissimilarity heat-map comparing plant-species composition across habitats. Values range from 0 (identical assemblages) to 1 (completely dissimilar), with warmer colours indicating greater compositional divergence. Each cell reports the pairwise Bray–Curtis distance between study sites.

Figure S2. Cumulative pollinator species richness per visit across study sites for each method: direct observations (blue), Automatic Camera System (orange) and both methods merged (black). The curves show how species richness increases with sampling effort, allowing comparison of completeness between focal observations.

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